



Geochemistry and petrogenesis of basaltic rocks from the Develidağ volcanic complex, Central Anatolia, Turkey

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ABSTRACT

The Develidağ volcanic zone (DVZ) is a part of the Middle Miocene-Quaternary Cappadocian Volcanic Province (CVP) in central Anatolia. The DVZ is located at the eastern side of the Plio-Quaternary Sultan-sazlığı pull-apart basin, that opened along the Ecemiş left-lateral strike-slip fault. Towards the south, the volcanic rocks of the DVZ overlie the Paleozoic metamorphic rocks of the Taurus range. Regional stratigraphic studies indicate that volcanic rocks of the DVZ crop out in a ~N–S trending rectangular-shaped exposure in the middle of which Middle Miocene andesitic rocks are found surrounded by Upper Miocene basaltic rocks. The purpose of this paper is to present basic geochemical data for the DVZ rocks and discuss possible processes of magma generation.

The Develidağ basalts are characterized by low LILE (Rb, K, Ba, Th) and high HFSE (Nb, Zr, Hf, Y) contents, whereas the andesites generally have high LILE and HFSE values except for Nb and Zr. Variable abundances of Pb (3.11–12.09 ppm) and U (0.36–2.64 ppm) are associated with high Ba content within the rock suites. Although low Nb/La (0.6–0.7) and relatively high Ba/Nb ratios indicate crustal involvement for the basalts, high Zr/Ba (0.5), Zr/Hf (42–47) and Th/U (3.13–4.69) values imply contributions from an asthenospheric source component. Furthermore, the high Zr/Hf values (>36) are the diagnostic feature of metasomatized mantle (Dupuy et al., 1992; Rudnick et al., 1993). Moreover, multi-element patterns show that Develidağ basalts have similar trace element signatures to those of the US Cascades tholeiites.

The ratios of Zr/Hf, Zr/Ba, Nb/Th and Sr/Ce indicate that basaltic rocks are derived from a MORB-like mantle, and calculated melting model reflects generation from a spinel peridotite source (3–4% melting), but the combined effects of melting and assimilation and fractional crystallization (AFC) processes seem to be partially responsible for the relatively evolved rocks. Typical tholeiitic-calk-alkaline associations are also observed among the volcanic rocks as a consequence of melting processes and crustal involvement, or due to the changes in the melting rate and degree. The lack of alkaline rocks that is conventionally observed during the final stage of many central Anatolian volcanic centers may be attributed either to the lack of extensional development or to a deficient amount of extension before the late Miocene.

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1. Introduction

Widespread volcanic activity developed on the Anatolian platform following the protracted continental collision between the Afro-Arabian and Eurasian plates that began during the early to middle Miocene (Şengör, 1980; Şengör and Yılmaz, 1981; Dewey et al., 1986; Fig. 1). Extensive volcanic activity was generated as a consequence of this collision and also by the effects of the previous subduction event. Although, continental collision led to progressive volcanism on all parts of the Anatolian block, the internal dynamics of the lithospheric faults play a critical role in controlling magmatic activity and previous studies record distinct types of volcanic activity in central Anatolia (Fig. 2); some are re-

lated to the subduction event (Pasquarè, 1968; Innocenti et al., 1975) whereas others reflect both subsequent and contemporaneous intraplate processes (Aydar et al., 1995; Notsu et al., 1995; Kürkcüoğlu et al., 1998, 2001, 2004).

The origin and evolution of magmatism on the Anatolian plate is a complex question given the dynamic nature of the tectonic regime. Extensional tectonism is primarily responsible for the magmatic activity in the Aegean portion of western Anatolia, while the eastern part of the Anatolian plate is governed by collisional tectonics. Magmatism in central Anatolia reflects the combined effects of these tectonic processes and is further constrained by movement along two lithospheric faults. Two voluminous strato-volcanoes (Hasandağ and Melendiz) and several monogenetic vents (e.g., Karapınar, Karacadağ) are located along the NW-SE oriented Tuz Gölü fault (Fig. 2), while many volcanic centers (e.g., Erciyes, Koçdağ, Alidağ) are situated between two segments of NE-SW oriented Ecemiş fault zone (Fig. 2).

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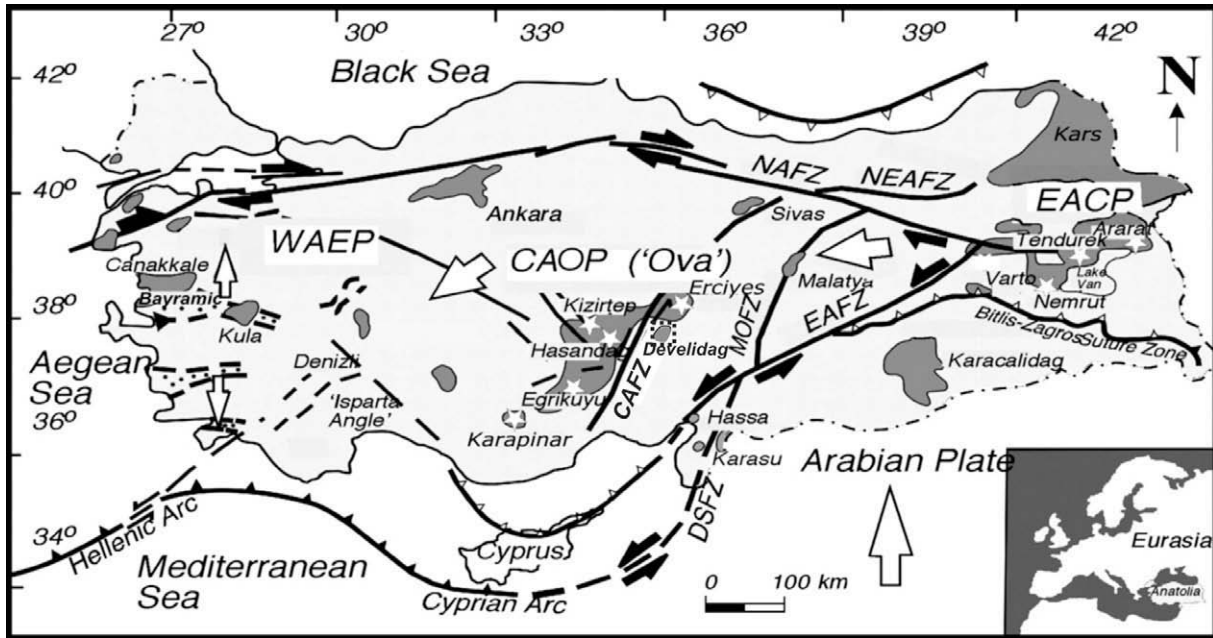


Fig. 1. Neotectonic map of Anatolia, indicating major fault systems and Plio-Quaternary volcanic areas (from Çoban (2007), modified from Innocenti et al. (1982), Kocyigit (1991), Kocyigit and Beyhan (1998), Yilmaz (1998) and Bozkurt (2001)). "Turkish escape" occurs along major strike-slip faults as indicated by arrows. DSFZ: Dead Sea Fracture Zone, EAFZ: East Anatolian Fault Zone, NAFZ: North Anatolian Fault Zone. Volcanic areas are shaded; stars indicate explosive volcanoes. CAOP: Central Anatolia 'Ova' Province, WAEP: Western Anatolia Extensional Province, EACP: East Anatolian Contractional Province, NEAFZ: North East Anatolian Fault Zone, MOFZ: Malatya-Ovacik Fault Zone. Study area is indicated with black dots.

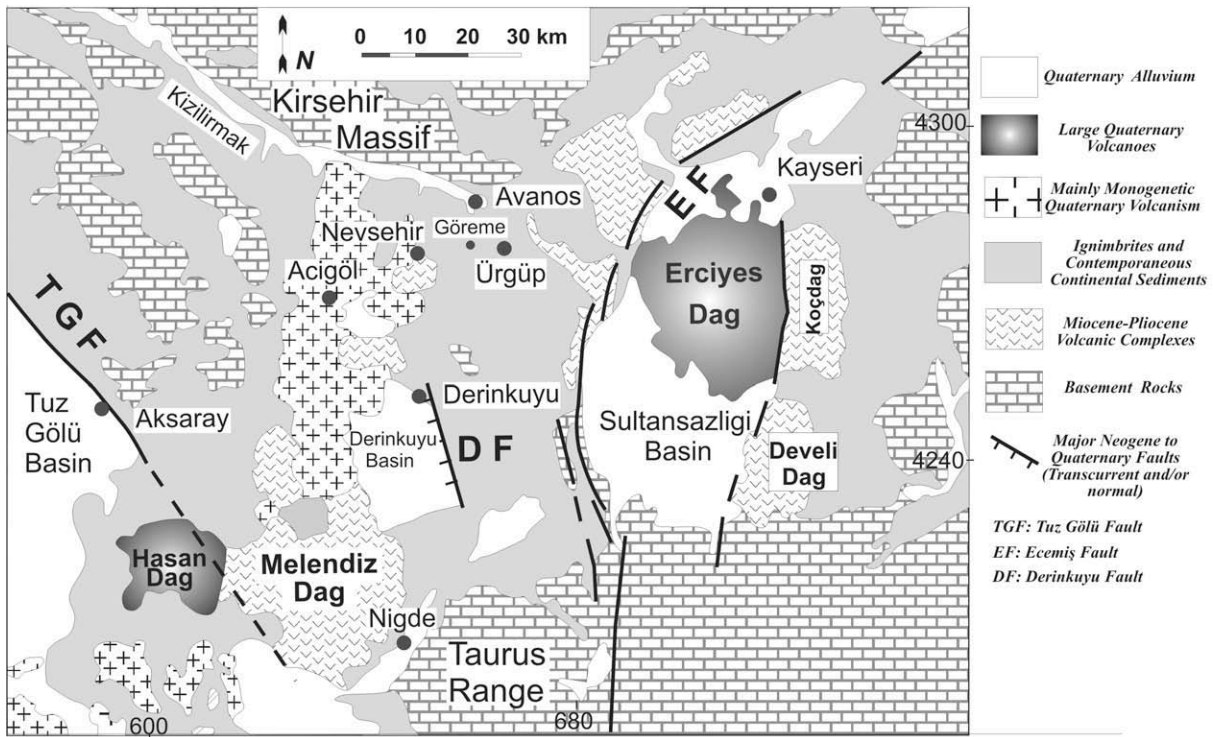


Fig. 2. Geological sketch map of the studied area adapted from Temel et al. (1998) modified from Pasquare et al. (1988) and Le Pennec et al. (1994).

Although, large volcanic provinces (Cappadocian volcanism) and individual stratovolcanoes (e.g., Hasandağ, Erciyes) in central Anatolia have been well-studied, relatively small volcanic centers such as Develidağ (Fig. 2) have not been investigated in depth so far. There are no prior geochemical data from this area (major and

trace elements, isotopic data or age determinations). The aim of this study is to use the geochemical data from young Develidağ lavas to investigate the source region for volcanism, and to give a plausible model for the genesis of the basaltic rocks. Investigations on Develidağ lavas will also contribute to the efforts on

studying geochemically underestimated monogenetic structures compared with the Strato-volcanoes. Furthermore, increasing number of data and studies on such volcanic vents will provide new insights for the better understanding of the geodynamics of Anatolia.

2. Geological background

The structural consequences of Afro-Arabian and Eurasian continental collision (Şengör, 1980; Şengör and Yılmaz, 1981; Dewey et al., 1986) are represented by deformation zones and pull-apart basins in Anatolian plate, one of which – the Sultansazlığı pull-apart basin – is situated in central Anatolia. The Miocene Develidağ volcanic complex is situated at the southeastern end of the Sultansazlığı pull-apart basin and the southern part of the eastern Ecemiş fault (Fig. 2); the western part of the volcanic center is bordered by the same strike-slip fault system (Koçyiğit and Beyhan, 1998). Previous structural studies suggest that Develidağ volcanism was generated prior to and during the extension of the Sultansazlığı pull-apart basin (Koçyiğit and Beyhan, 1998; Toprak, 1998). The regional geological map (Kayseri district) and superposed digital elevation model (Fig. 3) demonstrate clearly that basaltic and andesitic rocks cover significant surface and overlie the Paleozoic metamorphic rocks.

3. Petrographic descriptions

Basaltic volcanic rocks are collected from the freshest outcrops in the field but all are more or less altered especially in the groundmass. Basalts and basaltic rocks are generally represented by sheet-like plagioclases, rounded-olivines associated with the minor amount of pyroxenes. The groundmass consists of microlites of plagioclases, rounded forms of mafic phases and widely distributed opaques minerals. Phenocrysts of plagioclase, olivine and pyroxene reflect porphyritic and subophitic textures and comprise ~40 vol.% of these rocks. Plagioclases is the dominant phase in andesites and accompanied by the minor amount of pyroxenes. The groundmass also displays porphyritic texture and consists of plagioclase microlites, olivines and opaque minerals. These lavas contain roughly 25–30 vol.% phenocrysts.

4. Geochemistry

4.1. Sampling and analytical techniques

Eighteen samples were collected systematically to represent the range of rock types in the volcanic complex. Samples were cut and powdered using a rock saw and tungsten carbide disc mill, respectively, after removing the altered parts. Major, trace and rare earth element (REE) analysis (Table 1) were performed using both Directly Coupled Plasma (DCP) and Inductively Coupled Plasma-Mass Spectrometry (ICP-MS) on representative powders at Duke University, North Carolina, USA. Precision based on replicate analyses of samples and natural basalt standards is generally <1% for SiO₂, Sr, Y, Zr, Nb, La, Ce; <3% for other major elements, Ba, Sr, Rb, Cs, Cr, Sc, V, Co, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Hf and Ta; <5% Ni, Yb, Lu, Pb, Th and U. All oxides are presented as wt.%; trace elements are given in ppm.

4.2. Major element geochemistry

Develidağ volcanic rocks include both tholeiitic and calc-alkaline rocks. The great majority of the samples are basalt and basaltic andesite while a few samples are classified as andesite (Fig. 4). Basalts are mildly alkaline while remaining volcanic rocks plot below the alkaline – subalkaline division line of Miyashiro (1978).

Abundances of the major element oxides MgO, CaO, Fe₂O₃ and TiO₂ are linearly and negatively correlated with SiO₂, whereas K₂O contents show the opposite trend. These trends are expected in a system dominated by fractional crystallization (Fig. 5) although the nearly flat trends are observed for P₂O₅ and Na₂O. Linear negative trends observed between major element oxides and silica indicate the separation of ferromagnesian phases from the liquid, and the weak positive correlation of Al₂O₃ with SiO₂ (Fig. 5) suggests progressive removal of small volumes of plagioclase feldspar.

4.3. Trace and rare earth element geochemistry

Ni, Cr and V contents of the basalts range between 84 and 113, 67 and 129, 166 and 186 ppm, respectively whereas a few basalts and andesitic rocks have low Ni, Cr and V contents

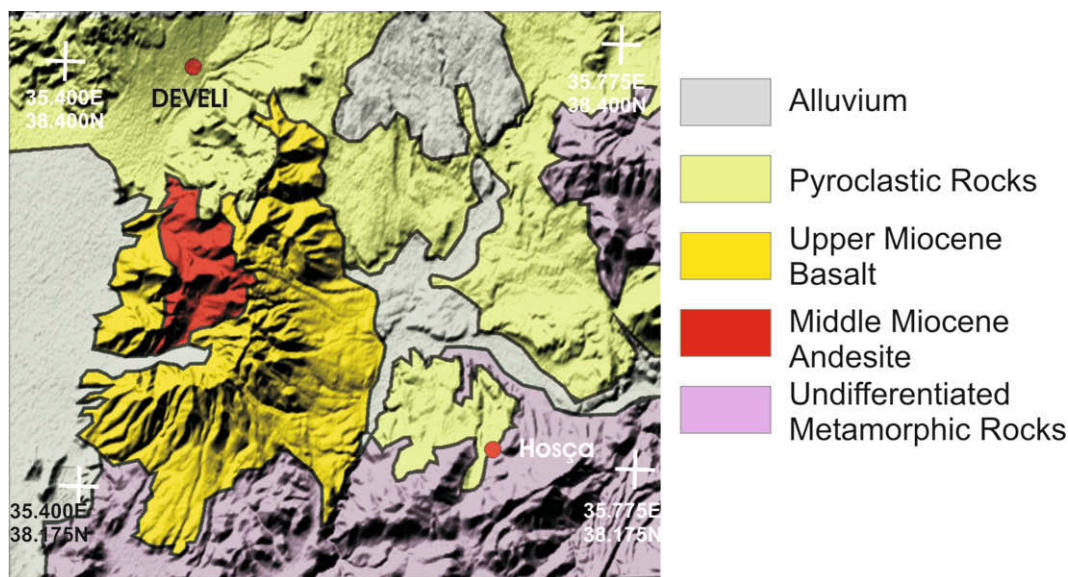
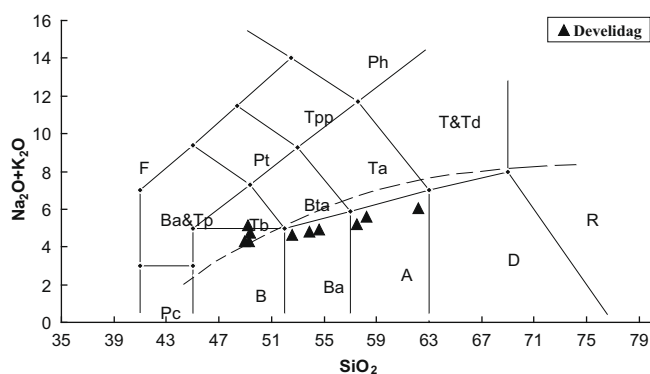


Fig. 3. Digital elevation model (DEM) and overlain simplified geological map of the study region (map adapted and redrawn from 1/5,00,000 Kayseri Section, General Directorate of Mineral Research and Exploration, Turkey).

Table 1

Major, trace and the rare earth element analysis of representative samples from Develidağ volcanic complex.

Sample	D2004-1	D2004-3	D2004-4	D2004-5	D2004-7	D2004-8	D2004-10	D2004-11	D2004-12	D2004-13
SiO ₂ (%)	49.24	49.36	57.46	58.24	54.64	53.86	49.32	48.95	52.59	62.19
Al ₂ O ₃	17.61	17.67	18.15	16.61	18.01	18.72	17.67	17.37	18.70	18.26
MgO	6.65	6.62	2.99	4.28	4.25	4.65	7.01	6.91	4.19	2.98
CaO	8.45	8.66	7.15	7.25	8.19	8.33	9.43	9.37	8.40	6.41
MnO	0.17	0.17	0.10	0.10	0.13	0.13	0.16	0.17	0.14	0.10
TiO ₂	1.84	1.83	0.85	0.79	1.11	0.92	1.56	1.57	1.15	0.66
Fe ₂ O ₃	11.19	11.17	6.77	6.32	7.97	7.92	10.57	10.61	8.87	5.93
Na ₂ O	4.23	4.16	3.87	3.78	3.89	3.85	3.66	3.63	3.76	4.16
K ₂ O	0.93	0.62	1.35	1.81	1.04	0.97	0.63	0.65	0.89	1.88
P ₂ O ₅	0.41	0.37	0.18	0.18	0.23	0.20	0.29	0.29	0.22	0.24
Total	100.3	100.2	98.7	99.1	99.2	99.4	100.0	99.2	98.7	102.5
Li (ppm)	6.98	7.60	13.17	12.48	6.24	8.95	8.79	8.00	7.52	9.13
Be	1.80	1.66	1.06	1.36	1.15	1.03	1.43	1.37	0.96	1.26
B	2.43	4.10	9.91	13.50	6.33	6.49	3.12	1.78	9.87	14.66
Sc	25.00	24.45	15.38	17.96	20.30	20.53	26.50	25.57	20.57	11.78
V	179.88	166.24	153.71	147.14	162.32	185.78	186.70	178.03	202.53	109.89
Cr	70.90	67.65	6.04	90.72	60.76	25.29	129.70	126.06	22.64	19.51
Co	56.72	61.47	30.52	41.26	113.85	43.78	59.72	63.11	46.48	40.32
Ni	86.15	84.11	0.00	32.61	19.61	6.97	113.70	110.38	15.40	5.21
Cu	38.92	57.61	14.23	40.40	33.40	36.73	64.49	55.35	48.07	17.64
Zn	112.22	84.94	57.23	61.45	73.11	61.68	76.39	77.28	72.06	68.20
Rb	6.80	6.17	36.39	54.67	21.74	17.64	4.65	4.93	11.97	34.58
Sr	433.62	428.53	497.27	472.14	467.77	517.20	463.31	447.35	521.08	475.96
Y	34.13	32.58	17.05	21.53	21.33	21.62	27.90	27.40	25.77	20.47
Zr	242.22	228.22	125.08	162.66	156.89	113.77	181.15	176.45	160.89	202.49
Nb	12.79	11.89	7.55	10.70	7.58	6.69	8.78	8.59	7.43	9.55
Mo	1.27	1.16	1.38	1.86	1.44	1.12	0.95	0.93	1.16	1.70
Sn	2.27	1.84	0.96	1.33	1.05	1.04	1.56	1.55	1.23	1.13
Sb	0.14	0.09	0.35	0.99	0.21	0.18	0.09	0.08	0.22	0.26
Cs	0.09	0.07	0.68	1.48	0.70	0.30	0.05	0.03	0.36	1.10
Ba	170.62	301.97	559.88	976.08	423.30	479.84	366.47	189.00	1528.54	550.95
La	18.33	17.43	19.29	24.63	16.35	16.01	13.95	13.86	13.96	21.78
Ce	41.84	39.72	34.40	42.78	32.67	30.86	31.54	31.34	30.01	42.21
Pr	5.62	5.36	4.41	5.36	4.02	3.82	4.19	4.16	3.97	5.20
Nd	23.20	22.03	16.67	19.81	15.85	15.17	17.55	17.30	16.73	19.34
Sm	5.48	5.28	3.50	4.06	3.59	3.45	4.31	4.21	4.07	3.92
Eu	1.73	1.72	1.15	1.20	1.27	1.13	1.49	1.47	1.58	1.27
Gd	5.21	4.97	3.29	4.03	3.50	3.46	4.22	4.14	4.31	3.70
Tb	0.99	0.96	0.52	0.62	0.61	0.59	0.80	0.78	0.73	0.60
Dy	6.27	6.06	3.11	3.70	3.85	3.75	5.12	5.05	4.65	3.63
Ho	1.34	1.31	0.63	0.77	0.82	0.80	1.10	1.09	1.00	0.77
Er	3.39	3.24	1.65	2.01	2.12	2.10	2.82	2.76	2.56	2.01
Yb	3.63	3.49	1.73	2.13	2.29	2.26	2.98	2.94	2.74	2.16
Lu	0.50	0.49	0.25	0.31	0.33	0.33	0.41	0.41	0.39	0.33
Hf	5.10	4.94	3.80	4.44	4.14	3.56	4.22	4.18	4.31	4.89
Ta	0.90	0.94	0.65	0.87	0.89	0.58	0.81	0.73	0.62	0.95
Hg	41.49	65.96	49.05	51.69	104.93	62.73	74.07	60.41	59.55	102.55
Pb	3.37	3.11	6.15	11.79	7.90	8.84	2.64	2.70	5.75	9.14
Th	1.27	1.16	6.26	9.64	4.73	3.27	1.52	1.51	2.63	6.39
U	0.40	0.36	1.66	2.64	1.19	0.90	0.35	0.32	0.69	1.54

**Fig. 4.** Total alkalis-silica diagram (Le Bas et al., 1986) showing Develidağ volcanic rocks. Alkaline-subalkaline division line from Miyashiro (1978).

(Table 1). Basalts have low Rb (4.7–6.8 ppm), moderate to low Nb (8.6–12.8 ppm) and high Zr values (1765–242 ppm), but

the relatively evolved rocks have highly variable elemental abundances (Table 1).

The abundances of highly incompatible large ion lithophile elements (LILE) Rb, Ba, Th, U, Pb and rare earth elements (REE, e.g., Ce) correlate positively with increasing SiO₂, but weak negative trends are observed for the high field strength elements (HFSE) Zr and Nb. (Fig. 5) These negative trends cannot be explained by simple fractional crystallization of the observed phenocryst phases, given the low bulk distribution coefficients for Zr and Nb. Sr contents weakly increase with increasing SiO₂ and decreasing MgO contents.

Primitive mantle-normalized spidergrams (Fig. 6a and b) indicate that basalts and basaltic andesites of the Develidağ volcanics have similar trace element signatures to those of the Columbia River tholeiites, Steens Mountain flood basalt, Rio Grande Rift axis rocks and Lassen region calc-alkaline basalts from Cascades range, USA (Hooper, 1988; Camp et al., 2003; Bacon et al., 1997; Duncker et al., 1991). In addition, similar incompatible element patterns (Fig. 6) are observed between andesitic rocks and calc-alkaline andesites from eastern and central Anatolia (Pearce et al., 1990;

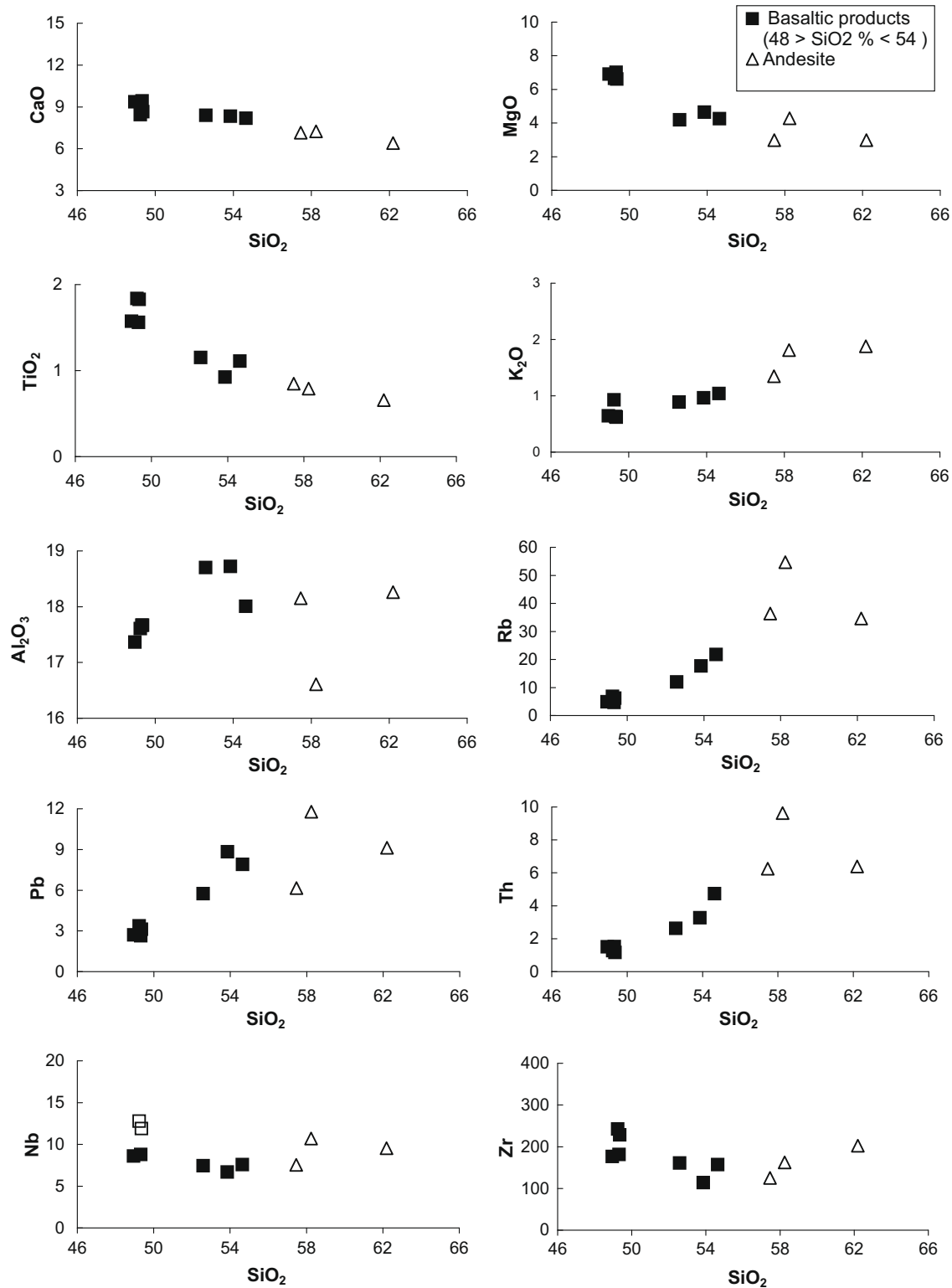


Fig. 5. SiO_2 plotted against the selected major and trace element data for Develidağ volcanic rocks.

Deniel et al., 1998) and Central America (El Salvador, Rotolo and Castorina, 1998).

Chondrite-normalized (Nakamura, 1974) REE patterns (Fig. 7) of all the Develidağ samples show weak LREE enrichment; all samples lack Eu anomalies. The Develidağ basalts have REE patterns similar to those of Columbia River and Rio Grande rift tholeiites and Cascades Range calc-alkaline basalts.

Specific trace element ratios can be used to investigate different geochemical processes and source components. For example, Hasse et al. (2000) suggested that low Nb/La (<1.3) implies crustal involvement, and Fitton et al. (1988) showed that high Ba/Nb ratios (>28) are typical of arc volcanism. These ratios are attributed to retention of Nb in rutile in shallow depths (<80 km). Menzies et al. (1991) have argued that Zr/Ba values can be used to distin-

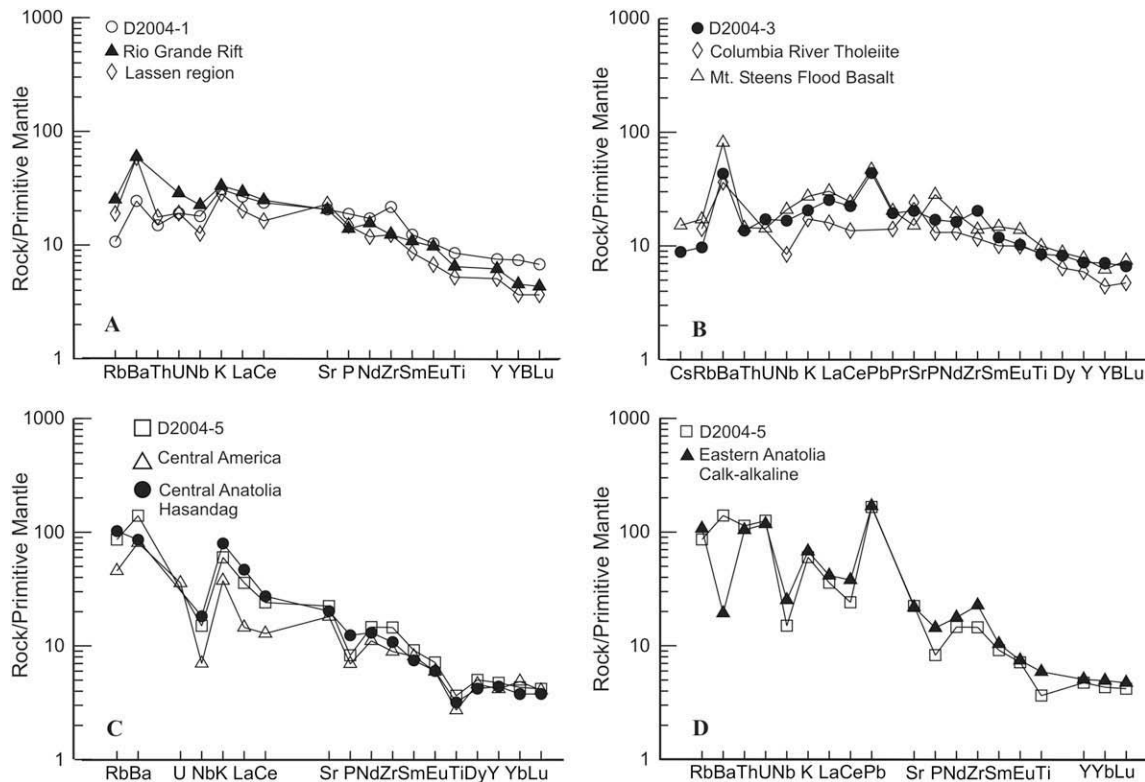


Fig. 6. Primitive mantle normalized (Sun and Mc Donough, 1989) multi-element patterns of the rock samples. Data sources from Camp et al. (2003), Deniel et al. (1998), Rotolo and Castorina (1998), Borg et al. (1997), Pearce et al. (1990), Duncker et al. (1991) and Hooper (1988).

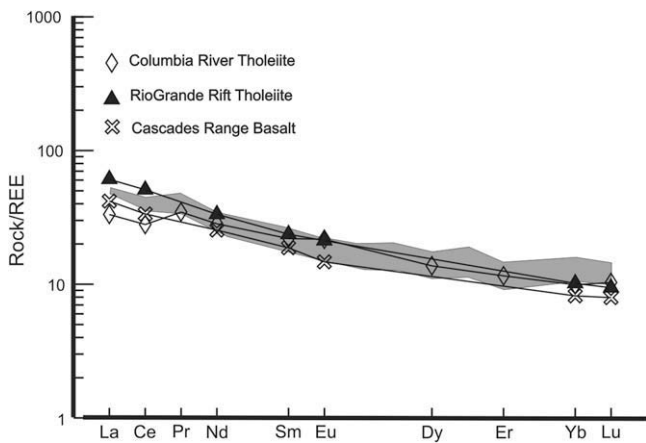


Fig. 7. Chondrite normalized (Nakamura, 1974) rare earth element patterns of the volcanic rocks. Data from Borg et al. (1997), Duncker et al. (1991) and Hooper (1988). Shaded region represents the basaltic rocks of the Develidağ volcanics.

guish lithospheric (0.3–0.5) from asthenospheric sources (Zr/Ba 0.5–0.7). The Develidağ complex basalts have low Nb/La (0.41–0.69) and moderate Ba/Nb (13–55) values, but their Zr/Ba ratios (0.49–1.41) suggest generation from the lithospheric sources.

Even though the trace element ratios shown above suggest crustal and lithospheric contributions dominate the genesis of volcanic rocks, some other incompatible element ratios indicate that the basaltic rocks may be derived from primitive sources. For example, the elevated Zr/Hf (42–47) ratios suggest (>36 is the diagnostic feature of metasomatized mantle Dupuy et al., 1992; Rudnick et al., 1993) that small-volume metasomatic fluids could enrich the source region (Furman and Graham, 1999) and the majority of the Develidağ samples have higher Zr/Hf ratios (42–

47) than this primitive values. Moreover, Th/U (3.13–4.69), Nb/Th (1.1–10.24) ratios are similar to OIB and Sr/Ce (10.36–14.68) ratios are similar to N-MORB values of Sun and Mc Donough (1989).

It is interesting to note that variable amounts of Pb (3–12 ppm) and U (0.36–2.64 ppm) are observed in samples that have high Ba contents. If high Ba values had resulted from lithospheric contributions, all high-Ba samples would have high Pb contents and LIL values. Therefore, these observed differences may instead result from heterogeneous chemical composition in the source region, or changes in melting degree or post-crystallization alteration, rather than the direct lithospheric involvement.

5. Discussion

Basaltic lavas are conventionally derived by mantle melting at depths shallower than about 100 km. The specific source regions contributing to basalt volcanism, however, may be located within either the lithosphere or the asthenosphere, and may represent a mixture of geochemical contributions from processes occurring over tens of millions of years or more. In this section, the chemical features of the Develidağ volcanics are used both directly and through modeling to identify the source characteristics and to determine the genesis of the rock suite.

5.1. Source implications

Both tholeiitic and calc-alkaline basaltic lavas erupted during the evolution of the Develidağ volcanic rock suite, together with minor amounts of andesite. Basaltic rocks are generally characterized by high Zr/Nb (17–21), Ba/Nb (13–205), moderately high La/Yb (4.7–7), low Nb/La (0.4–0.7) and moderately low Ce/Pb (12.7–3.4). These different ratios imply different geochemical processes that present an overall picture of great complexity. Ba/La–Nb/La (Fig. 8) and Ba/Nb–La/Nb (Fig. 9) variations show that the Devel-

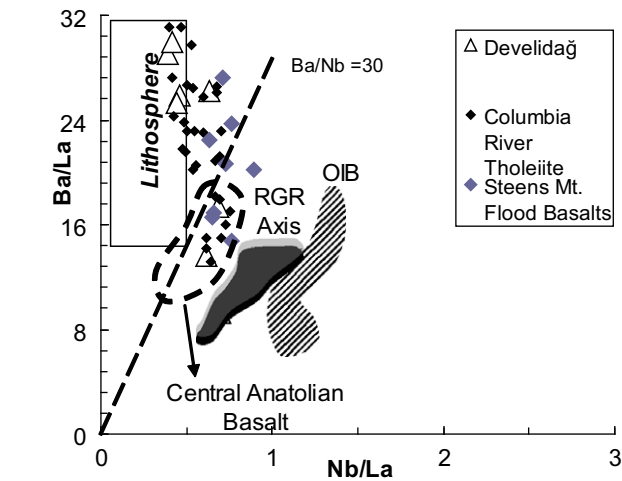


Fig. 8. Ba/La–Nb/La diagram of the Develidağ volcanic rocks. Lithosphere region is from Ormerod et al. (1991). Data sources from Camp et al. (2003), Kürkcüoğlu et al. (2001), Gibson et al. (1993), Chen et al. (1990) Hooper (1988) and Chen and Frey (1985).

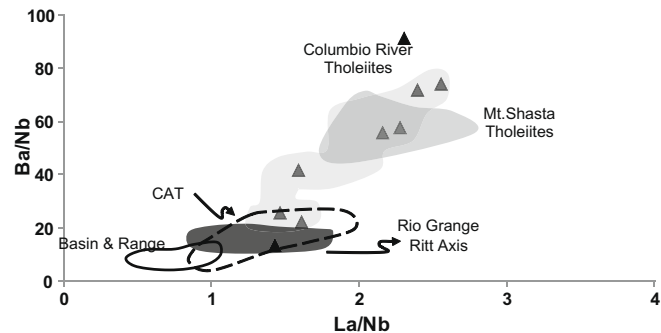


Fig. 9. Ba/Nb–La/Nb diagram of the volcanic rocks. Data sources from Kürkcüoğlu et al. (2001), Deniel et al. (1998), Bacon et al. (1997), Gibson et al. (1993), Kempton et al. (1991,1987) and Hooper (1988). CAT: Central Anatolian Tholeiites.

idağ samples plot within fields defined by Columbia River tholeiites (CRT; Hooper, 1988) and Mt. Shasta tholeiites from the Cascades. The geochemistry of Columbia River tholeiites indicates that these magmas formed by decompression melting in the tectonic setting of a back-arc basin (Hooper, 1988). The source regions contributing to Columbia River magmatism include both old sub-continental lithosphere and mantle peridotite that had been affected by aqueous solutions emanating from subducted sediment of the Juan de Fuca Plate (Faure, 2001). Recent studies suggest an alternative CRB model call for a mantle plume impinging the continental lithosphere at about 15–16 Ma when the Steens basalts erupted. The continental lithosphere in this region is the west of Paleoproterozoic–Archean Wyoming craton, and consists of pri-

Table 2
CIPW norms, mineral/melt partition (K_D), calculated whole-rock partition coefficients (D_0) and the trace element values of the starting composition used in the melting model. CIPW norms for the starting compositions are as follows: Ol: 62; Cpx:12; Opx:24; Spinel: 2 for spinel peridotite; Ol: 65; Cpx:3; Opx: 28; Garnet: 4 for garnet peridotite).

	Mineral-melt partition coefficients (K_D)				Calculated whole-rock bulk-partition coefficients (D_0)	Starting composition garnet peridotite (66SAL-1, Frey, 1980)
	Ol	Opx	Cpx	Garnet		
La	0.0067	0.02	0.056	0.001	0.0120	2.1
Sm	0.0066	0.05	0.5	0.290	0.0450	1.3
	Mineral-melt partition coefficients (K_D)				Calculated whole-rock bulk-partition coefficients (D_0)	Starting Composition spinel peridotite (McDonough, 1990)
	Ol	Opx	Cpx	Spinel		
La	0.0067	0.02	0.056	0.01	0.018	2.60
Sm	0.0066	0.05	0.5	0.0064	0.077	0.47

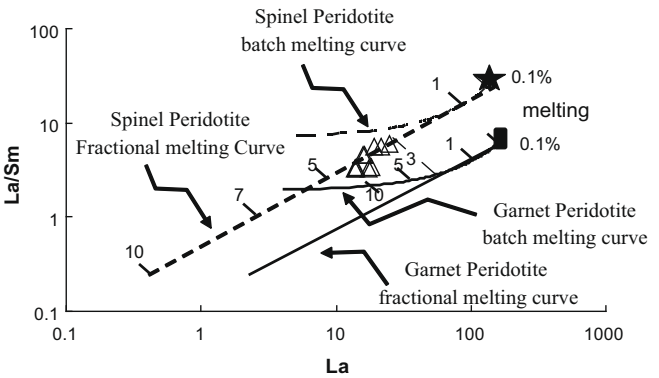


Fig. 10. Melting models of La/Sm–La variations starting from the spinel and the garnet peridotite source compositions. Starting compositions are from Sun and Mc Donough (1989; spinel) and Frey et al. (1980; garnet). Tick marks express the melting degrees.

Table 3
Calculated model trace element concentrations (ppm) via batch and fractional melting from the spinel and the garnet peridotite source composition. F = melting degree.

F (%)	0.1	1	3	5	7	10
Spinel peridotite batch melting						
La	139.92	94.81	55.23	38.97	30.10	22.44
Sm	6.01	5.43	4.48	3.81	3.31	2.77
Spinel peridotite fractional melting						
La	139.70	84.30	26.98	8.43	2.57	0.41
Sm	6.01	5.39	4.23	3.30	2.56	1.73
Garnet peridotite batch melting						
La	166.82	97.75	50.91	34.41	25.99	19.01
Sm	28.41	23.92	17.70	14.04	11.64	9.26
Garnet peridotite fractional melting						
La	166.24	76.89	13.51	2.29	0.37	0.02
Sm	28.41	23.42	15.16	9.72	6.18	3.07

marily Mesozoic accreted terrains (Hanan et al., 2008; Camp and Hanan, 2008; Camp et al., 2003). In addition, Baker et al. (1994) concluded that high-alumina olivine tholeiites at Mount Shasta represent 6–10% nearly anhydrous melts of depleted mantle that had previously been enriched with a modest amount of a subduction-related component (e.g., Donnely-Nolan et al., 1991). The geochemical similarities between the Cascades lavas and basaltic rocks of Develidağ suggest these suites can be derived from similar mantle source(s).

In order to investigate the mineralogical characteristics of the source region for the basaltic rocks, melting calculations were performed to evaluate the probable trace element contents at given melting degrees (F), starting from both garnet and spinel peridotite end members. Trace element concentrations and CIPW norms of the starting composition, mineral/melt partition coefficients (K_D), and calculated bulk rock distribution coefficients (D_0) are given

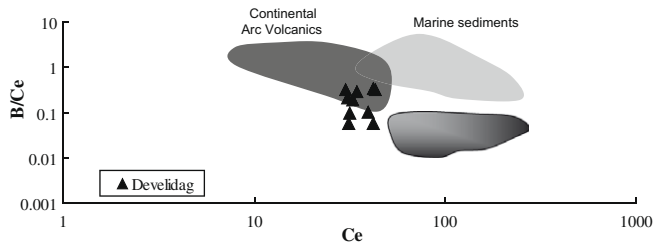


Fig. 11. B/Ce–Ce variations of the Develidağ volcanic samples. Data from Chan et al. (1999), Ryan et al. (1996) and Dean and Parduhn (1983).

in Table 2. Two of the samples plot close to the spinel peridotite fractional melting curve (Fig. 10), while rest of the basaltic rocks plot directly on the curve (Table 3). These models thus indicate that basaltic rocks of the Develidağ complex could be derived by fractional melting (3–4%) from a spinel peridotite source.

Moreover, B/Ce–Ce variations (Fig. 11) show that Develidağ basalts plot within and adjacent to the OIB field, while the calc-alkaline samples plot between the regions defined by OIB and continental arc volcanics. These figures and calculated melting model clearly imply that volcanic rocks were derived as a consequence of the interaction of at least two, possibly three different mantle and continental sources with the combined effect of the melting and combined assimilation / fractional crystallization in an extension-related region.

5.2. Assimilation of crustal material

The chemical composition of magma can be modified through assimilation of crustal materials during ascent and storage; these modifications may be indicated by several conventional incompatible trace element ratios diagnostic within the arc and subduction environment.

Basaltic rocks of the Develidağ complex have high Ba/Nb (>28) ratios which is a common feature of arc magmas (e.g., Fitton et al., 1991), and high La/Nb ratios (1.4–2.4) indicating crustal involvement (Hasse et al., 2000). Further, prevalent low Ce/Pb (3.5–12) and Rb/Ba values also suggest lithospheric contributions. Even though these geochemical indicators imply crustal contributions,

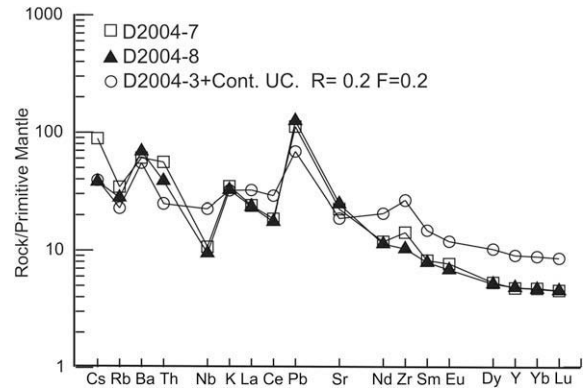


Fig. 12. Primitive mantle-normalized model trace element patterns, starting from the tholeiitic end member (D2004-3), calculated using the AFC equations of De Paolo (1981) and compared to andesitic rocks of Develidağ.

primitive-mantle-normalized spidergrams and rare earth element patterns (Figs. 6 and 7) do not reflect the contamination signature.

Relative to the basaltic rocks, andesites of Develidağ volcanics are represented by low Ce/Pb, Nb/U, Nb/Y and high La/Nb, La/Ta ratios. This tendency is also observed in multi-element patterns. Primitive mantle-normalized spidergrams display similar signature to those of Central (Hasandağ) and Eastern Anatolian samples and also to those of central American calc-alkaline rocks with negative Nb anomalies (Fig. 6c and d). Incompatible element ratios and spidergrams indicate that relatively evolved rocks (57–62 wt% SiO₂) of Develidağ volcanics could be more contaminated by crustal materials than the basaltic ones. In order to shed more light on the origin of the rock samples, AFC modeling (De Paolo, 1981) is performed to calculate the possible LIL and HFS element content, starting from the accepted contaminant and end-member.

5.2.1. AFC modeling

In order to shed more light to the origin of the andesitic rocks (57–62 wt% SiO₂), model trace element concentrations are calculated starting from the tholeiitic source component. For this purpose, sample 2004-3 is chosen as a starting composition, due to

Table 4

Calculated trace element concentrations (ppm) from the AFC modeling and some measured calc-alkaline samples^a.

	Calculated whole-rock partition coefficient (D_0)	Starting composition ^b	Contaminant ^c	Calculated trace element composition ($F = 0.2$ $r = 0.2$)	D2004-7	D2004-8
Rb	0.0687	6.17	112	14.49	21.74	17.64
Sr	1.2213	417.73	350	389.83	458.91	508.41
Ba	0.2704	301.97	550	383.1	423.3	479.84
Nb	0.1076	11.89	25	15.96	7.58	6.69
Zr	0.0490	228.22	190	293.2	156.89	113.77
K	0.1089	0.62%	2.80%	0.80%	1.04	0.97
Y	0.1157	32.58	22	40.34	21.33	21.62
La	0.2031	17.43	30	22.04	16.35	16.01
Ce	0.1938	39.72	64	50.92	32.67	30.86
Nd	0.2126	22.03	26	27.52	15.85	15.17
Sm	0.2298	5.28	4.5	6.46	3.59	3.45
Eu	0.3953	1.72	0.88	1.97	1.27	1.13
Gd	0.2476	4.97	3.8	6.02	3.5	3.46
Dy	0.1885	6.06	3.5	7.4	3.85	3.75
Er	0.3090	3.24	2.3	3.85	2.12	2.1
Yb	0.1871	3.49	2.2	4.27	2.29	2.26
Th	0.0123	1.16	10.7	2.11	4.73	3.27
Lu	0.0980	0.49	0.32	0.62	0.33	0.33
Pb	0.2284	3.11	20	4.85	7.9	8.84
Cs	0.1008	0.07	3.7	0.31	0.7	0.3

^a CIPW norms of the starting composition are as follows: Plg: 60.3; K-felds: 3.66; Ne: 1.42; Ol: 17.69; Cpx: 10.53.

^b D2004-3.

^c Upper Crust after Taylor and McLennan (1985).

being the most primitive one among the samples. The starting and contaminant composition and calculated D_0 values are given in Table 4. Primitive mantle-normalized trace element patterns show that basaltic andesites and andesites have similar trends to those of the model pattern (Fig. 12). LIL elements of the calculated model and measured samples have approximately the same values whereas HFS elements of the model are slightly higher than the measured values. Nb and Pb are depleted and enriched respectively as a consequence of crustal involvement. These patterns indicate that these rocks can be derived by approximately 20% assimilation of the crustal material(s) (Fig. 12) from a tholeiitic parent.

5.3. Origin of the volcanic rocks

Although, coexisting tholeiitic and calc-alkaline suites are not very frequent in volcanic settings, these associations are observed at several different locations in central Anatolia such as Erciyes (Kürkcüoğlu et al., 1998 and Kürkcüoğlu et al., 2001), Hasandağ (Aydar et al., 1995 and Deniel et al., 1998) and Sivas region (Furman et al., 2007). It is conventionally accepted that a tholeiitic parent could generate a calc-alkaline derivative magma(s), related to the melting process.

Trace element abundances, including REE patterns and incompatible element ratios, and diagnostic process diagrams imply that basaltic rocks and associated low SiO_2 evolved rocks are derived by the combined effect of the following processes:

- (i) *Partial melting* is responsible for the generation of the basalts and basaltic rocks reflecting 3–4% melting, from a depleted mantle, possibly from a MORB-like source
- (ii) *Assimilation and fractional crystallization*: samples with higher SiO_2 content (>52 wt.%) can be derived from a similar tholeiitic source with the combined effects of both partial melting and the AFC process. Model calculations suggest 20% assimilation of the crustal material during the ascent of magma from the melting region to the surface.

6. Conclusion

Central Anatolia is represented by abundant severe volcanic activities, including three main stratovolcanoes (Erciyes, Hasandağ and Melendiz), some monogenetic systems (Acigöl, Karadağ, Karacadağ) and other lava producing vents (Konya, Sivas), all situated within the central Anatolia. Volcanic activities in central Anatolia are related to the internal deformation via the two main lithospheric faults, Ececiş in the east and Tuzgölü in the west (Fig. 2). These fault systems are mainly responsible for the volcanic activities that developed in the central Anatolia. The common features of the central Anatolian volcanic rocks are abundant tholeiitic and calc-alkaline associations, followed by alkaline activities at the final stage of the volcanic systems. Develidağ volcanic system is represented by mainly tholeiitic basalts and associated with minor amount of calc-alkaline products. Melting model indicates that basaltic rocks can be derived by partial melting (3–4%) from a spinel peridotite source component whereas calc-alkaline rocks show minor amount of subsequent crustal contribution during differentiation as indicated by LIL/HFS element ratios and supported by AFC modeling. Development of the tholeiitic calc-alkaline associations at Develidağ volcanic system could be related to the activities of the eastern segment of the Ececiş fault. Tholeiitic and calc-alkaline associations can be explained either by combined development of partial melting of the mantle source and following crustal involvement or by the change in the rate of the melting degree. The lack of the alkaline rocks seems to be attributed to either

the lack of extensional development or to the deficient amount of extension in Central Anatolia before the late Miocene.

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